

THINKING LIKE A CEA EXPERT

An Intuition Guide
based on the following exercise:
[LINK TO CASE STUDY](#)

Bloomberg
Philanthropies



DATA FOR
HEALTH INITIATIVE



CDC Foundation
Together our impact is greater



VANDERBILT
Center for Health
Economic Modeling



Vital
Strategies

Case Study Background:

Several years of teaching workshops has led to a new illness being observed among health economists. As revealed by new research from the pioneering decision scientist, Dr. Grooves, this new disease is a consequence of the enduring sadness that results from the workshop being over. This condition has come to be called the Brief Yearning Episode (commonly abbreviated to BYE). Our team has exclusive access to data on the natural history of BYE.

These data were gathered from an observational study that followed a cohort of 30-year old students for 3 years. At the start of the study, 60% have Mild BYE and 40% were healthy.

BYE is associated with reduced quality of life. The utility weight for Mild BYE is 0.88 and the utility weight for Severe BYE is 0.65, whereas individuals without BYE have a utility of 1.0.

Everyone represented in the model faces age-specific background mortality, which can be based on the provided life table. Individuals with BYE have a risk of dying from sadness. Mild cases have a 1.4% annual probability of dying from sadness and severe cases have a 4.21% annual probability of dying.

The data also suggested that healthy individuals that survive background mortality have a 40.71% chance of developing BYE within a year of the workshop. Among individuals newly developing BYE, 80% have Mild BYE and the rest have Severe BYE. It has been estimated that 30% of the individuals with Mild progressed to Severe over a 3-year period. Individuals with Mild BYE can fully recover and return to the Well state, with an annual probability of 7%, conditional on surviving to the end of the year.

Patients with Severe BYE have a 10.5% annual probability of returning to the Mild state. Individuals with Severe BYE cannot return directly to the Well state. Individuals who recover from previous episodes of BYE face the same risk of developing the condition again as those who have not yet had BYE.

To treat BYE, the only solution is a beach holiday. For mild cases, a \$3,000 trip per year is required and for severe cases, a \$9,000 trip is required. It is recommended that patients moving from mild to severe, have a one-time treatment at progression of upgrading to a private jet for \$35,000.

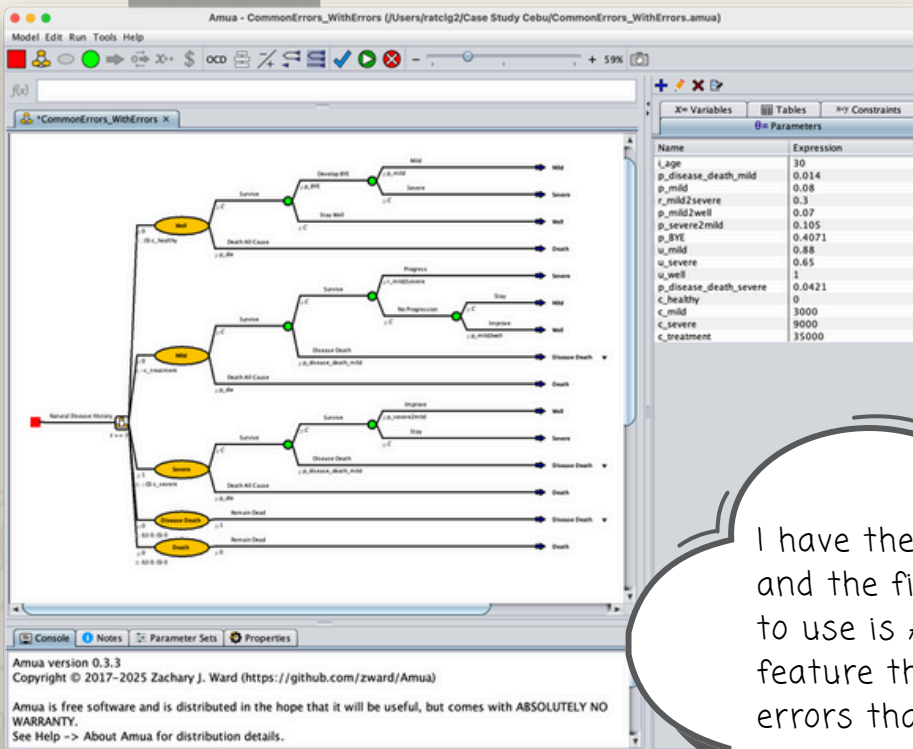
A similar disease CYA that is seen in economic workshops but has slightly more expensive treatment and slightly worse outcomes has been found to cost about \$100,00 per person and result in 10 QALYs.

Dr. Grooves develop a 5-state Markov model: (1) Well, (2) Mild, (3) Severe, (4) Disease Death and (5) [All Cause] Death based on the data provided above. He said he used a 3% discounting rate for both costs and utility and half-cycle correction on the model. All costs were converted into the same year. He is hoping to find the life-time impacts of BYE.

He also provided you some results from other papers that he think you could use to help understand your results. (Added later)

This Guide's Style:

This guide is written as if you are in the brain of the expert and walks you through their thoughts. The goal is to help you build intuition for doing quality assurance on your own model as well as reviewing external resources. This is heavily focused on Amua but many of the concepts can be applied to other softwares.



I have the model open in Amua and the first thing that I want to use is Amua's "Check Model" feature this will show me any errors that can be quickly fixed

Check The
Model

Okay so this tells us that me that the death node has probabilities that do not sum to 1. Let me look at that.

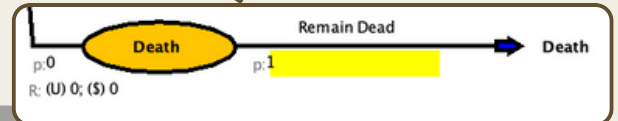
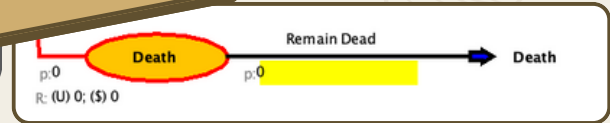
Console Notes Parameter Sets Properties

Treatment	23081.0559	1.614738517316E8	23081.0559	9.102
> 1 errors: Node Death: Probabilities sum to 0.0!				

Ah yes, I forgot to add the 100% probability that keeps all individuals in the "Death" end state. Well I am here, I also verified that the Death state has the end state of "Death"

Fix #1

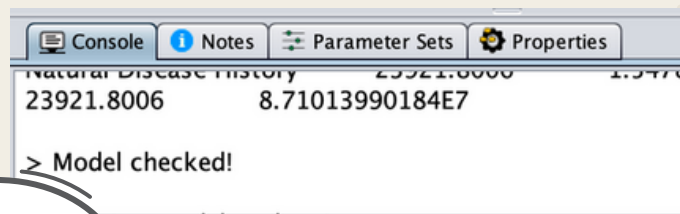
Set p:1 for the "Death" node



Remember that end states mean that no one can leave once they enter. Make sure there is an 100% chance that everyone stays in that state. (We don't zombies!)

Now that we have fixed that error, we can run the check again to see if there is anything else we missed.

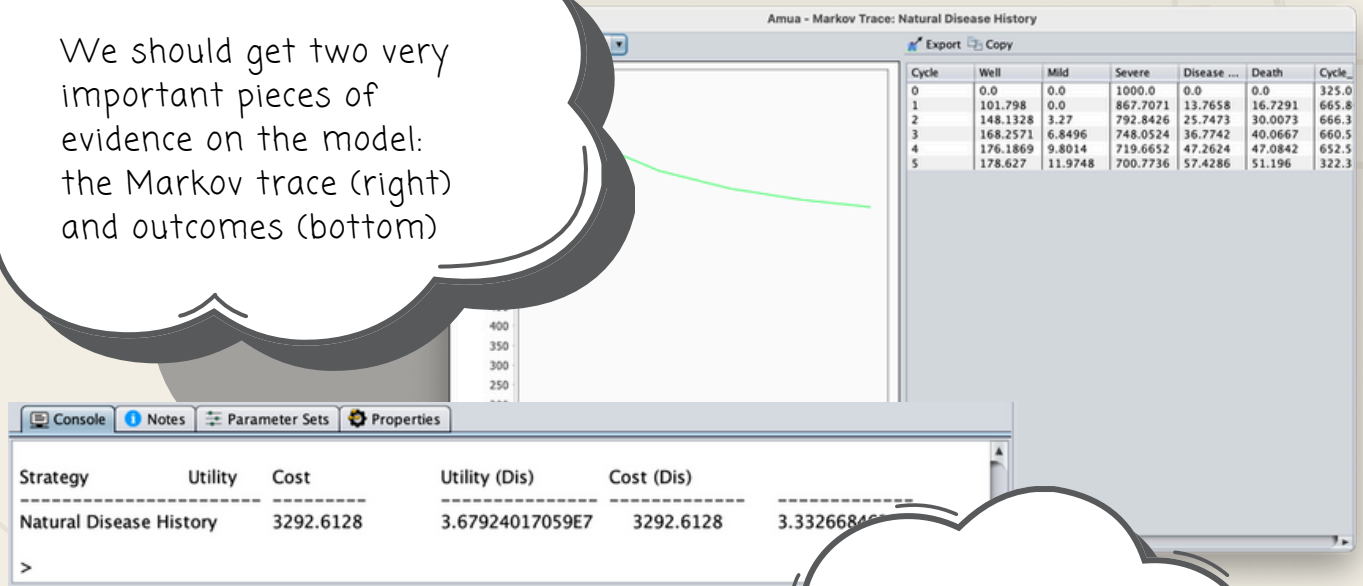
Check The
Model



Okay, perfect. The "Model checked!" notification in the Console means that the model has code that will run. It does not mean that I have a perfect model.

Run The
Model

We should get two very important pieces of evidence on the model: the Markov trace (right) and outcomes (bottom)



I want to start with the Markov trace because I already see an issue.

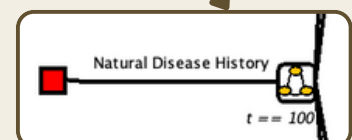
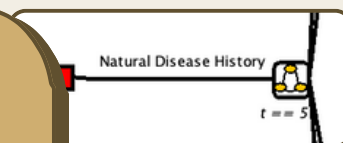
"He is hoping to find the life-time impacts of BYE."

Remember that a "Lifetime model runs until everyone dies. The number of cycles varies based on the initial age. You can over estimate but each additional cycle does add run time to model.

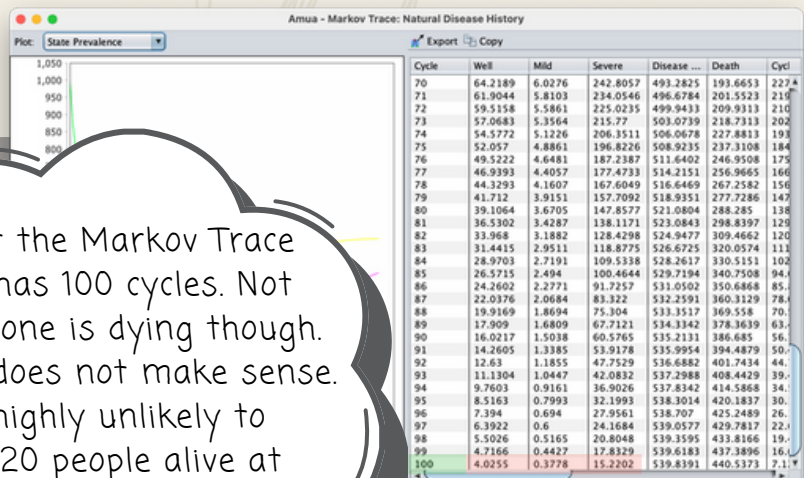
The model needs to be a lifetime model but this only has 5 cycles in the Markov Trace

Fix #2

Set $t=100$ for the terminal condition



Cycle	Well	Mild	Severe	Disease ...	Death	Cycle...
0	0.0	0.0	1000.0	0.0	0.0	325.0
1	101.798	0.0	867.7071	13.7658	16.7291	665.8
2	148.1328	3.27	792.8426	25.7473	30.0073	666.3
3	168.2571	6.8496	748.0524	36.7742	40.0667	660.5
4	176.1869	9.8014	719.6652	47.2624	47.0842	652.5
5	178.627	11.9748	700.7736	57.4286	51.196	322.3



Great the Markov Trace now has 100 cycles. Not everyone is dying though. This does not make sense. It is highly unlikely to have 20 people alive at 130 years old

First I want to check the structure of the model. We just fixed the Death node so let's check all branches that lead to death.

Well to Death:

- ☒ has branch
- ☒ p_die
- ☒ end state is Death

Severe to Death:

- ☒ has branch
- ☒ p_die
- ☒ end state is Death

Mild to Death:

- ☒ has branch
- ☒ p_die
- ☒ end state is Death

Okay the structure looks good, now I want to look at the table inputs

Amua - Define Table

Name: `tbl_p_die` Table Type: `Lookup` Interpolation Me...: `Linear`

Import Export Rows: `112` Cols: `2` Lookup Method: `Interpolate` Extrapolate: `No`

age	p_die
0.0	0.01672...
1.0	0.01369...
2.0	0.01065...
3.0	0.00760...
4.0	0.00454...
5.0	0.00146...
6.0	0.00129...
7.0	0.00112...
8.0	9.59497...
9.0	7.89451...
10.0	6.19377...
11.0	6.02151...
12.0	5.84925...
13.0	5.67699...
14.0	5.50472...
15.0	5.33245...
16.0	5.70833...
17.0	6.08419...
18.0	6.46004...
19.0	6.83587...
20.0	7.21169...
21.0	7.92186...
22.0	8.63107...

Notes:

Save Cancel

- ☒ Index Column
- ☒ Values for all indices
- ☒ Table type: Lookup
- ☒ Method: Interpolate

Not looking like a table issue. I will now go look at the variable for p_die.

Amua - Define Variable

Name: `p_die`

f(x) Expression: `tbl_p_die[t,1]`

Expected Value:

Notes:

Okay so:

- ☒ Correct table name
- ☒ Looking at the 1st column which is the p_die column
- ☐ Using Amua time variable "t" to look up the index

Amua - Define Parameter

Name: `Age`

f(x) Expression: `30`

Expected Value:

Notes:

WAIT! Our initial age is 30. This means that right now "t" is using the cycle number not the age of the person to index the table.

Fix #3

Change "t" to include the initial age (t+i_age)

Remember that the Amua built in "t" variable is the cycle number or time in the model. It will be 1 for cycle 1, 2 for cycle 2, and so on.

Amua

Name: `p_die`

f(x) Expression: `tbl_p_die[t,1]`

Expected Value:

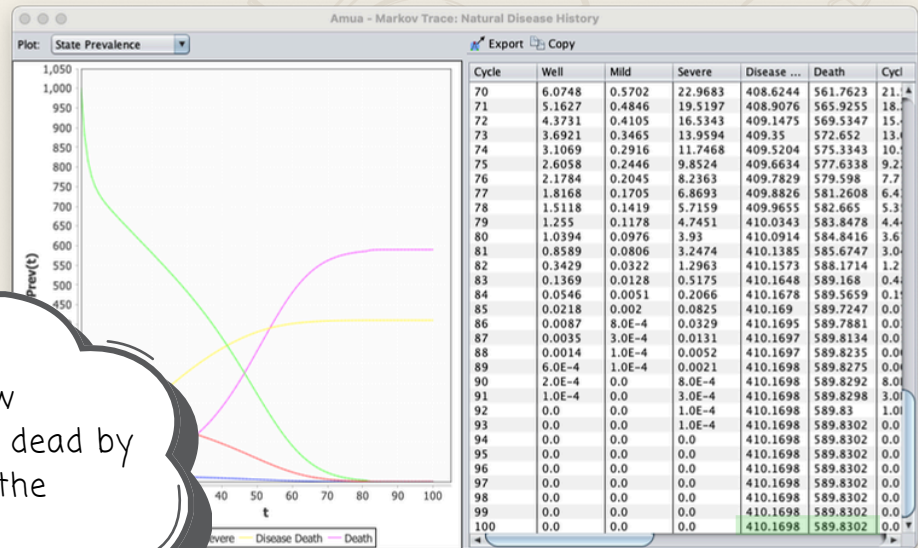
Amua

Name: `p_die`

f(x) Expression: `tbl_p_die[t+i_age,1]`

Expected Value:

Run The
Model Again



Perfect, now everyone is dead by the end of the model.

Now when I look at the graph I'm noticing something else.

"At the start of the study, 60% have Mild BYE and 40% were healthy."

At cycle 0, all 1,000 people are in the severe state.

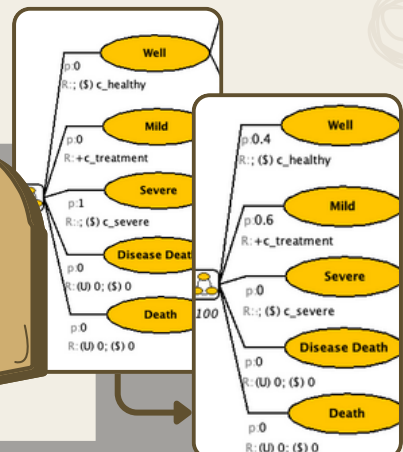
Remember the starting probabilities set the number of individuals that start in each state.

I need to check the starting probabilities.

The starting probabilities are completely different than what we wanted to model.

Fix #4

Set
p: 0.4 for Well
p: 0.6 for Mild
p: 0 for Severe



Run The Model Again



Great! Now we have individuals starting in Well or Mild.

Looking at the graph again, I notice there is a large spike in severe cases

Let me look at the background to see if this makes sense.

"Among individuals newly developing BYE, 80% have Mild BYE and the rest have Severe BYE."

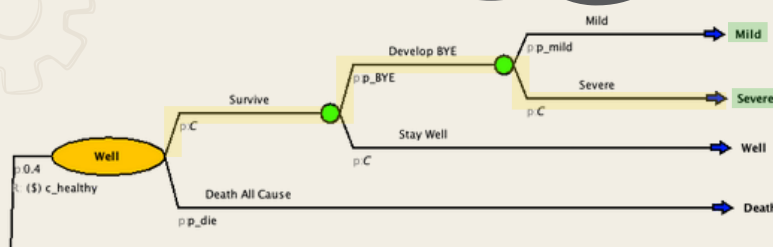
"30% of the individuals with Mild progressed to Severe over a 3-year period"

Based on this, I would expect Mild to spike first then to see Severe grow over time.

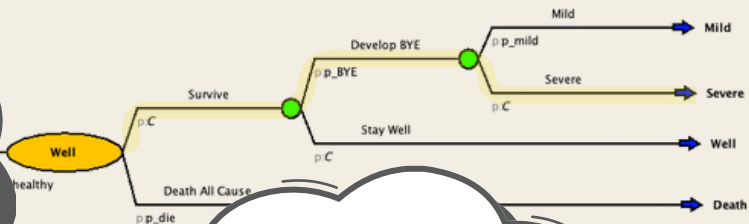
Let's start by looking at the structure of the model. I should start with Well or Mild since everyone starts in these states

I'm going to look at the Well state, specifically the flow to Severe.

Looking at this I can see that the Well state correctly has Mild and Severe end states so that is not the issue.



Now I want to check the probability parameters.



I just fixed p_{die} so I know that is correct. and the Survive has the "C".

I will start with p_{BYE}

"...healthy individuals that survive background mortality have a 40.71% chance of developing BYE within a year of the workshop

Since we have annual cycles. I know that p_{BYE} should be 0.4071.

Name	Expression
i_age	30
p_disease_death_mild	0.014
p_mild	0.08
r_mild2severe	0.3
p_mild2well	0.07
p_severe2mild	0.105
p_BYE	0.4071
u_mild	0.88
u_severe	0.65

Great it is correct!

Now let me check p_{mild}

Name	Expression
i_age	30
p_disease_death_mild	0.014
p_mild	0.08
r_mild2severe	0.3
p_mild2well	0.07
p_severe2mild	0.105
p_BYE	0.4071
u_mild	0.88
u_severe	0.65

Oh no, a typo!

Based on the background $p_{mild} = 0.8$

"...developing BYE, 80% have Mild BYE and the rest have Severe BYE."

Fix #5

$p_{mild} = 0.8$

Name:

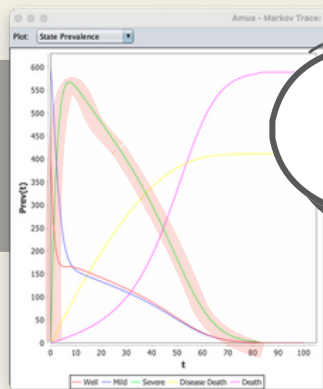
f(x) Expression: 0.08

Name:

f(x) Expression: 0.8



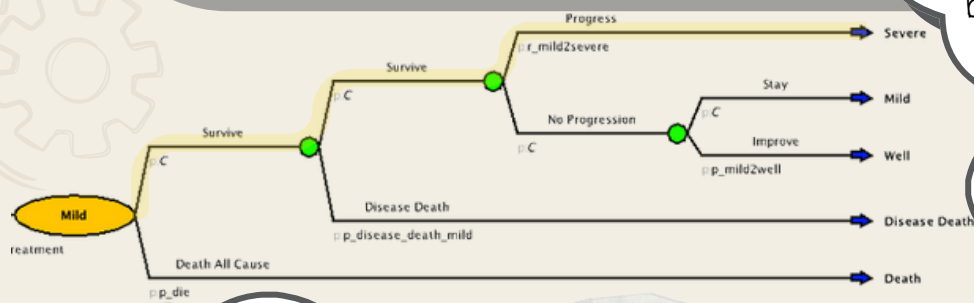
I finished checking the Well state so now I will check the Mild state specifically the flow to Severe.



The spike in Severe got lower but it still feels high to me. I am going to do additional checks

I see that the Mild state has the correct end state for each branch.

Now I want to check the probability parameters.



`p_disease_death_mild` is correct

"Mild cases have a 1.4% annual probability of dying"

Name	Expression
<code>i_age</code>	30
<code>p_disease_death_mild</code>	0.014
<code>p_mild</code>	0.8
<code>r_mild2severe</code>	0.3

"30% of the individuals with Mild progressed to Severe over a 3-year period"

This is a 3-year rate. I need to make sure this is an annual probability in the model

Name	Expression
<code>i_age</code>	30
<code>p_disease_death_mild</code>	0.014
<code>p_mild</code>	0.8
<code>r_mild2severe</code>	0.3

Yikes! this was entered into the model as a rate.

Remember that you cannot mix rates and probabilities. Additionally, all probabilities must be for the cycle length.

Fix #6

- 1.Delete rate
- 2.Add probability
- 3.Update model

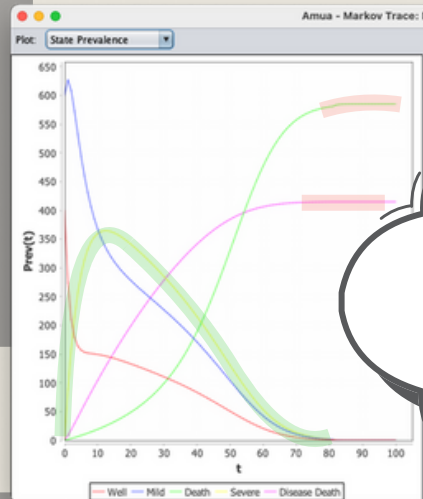
Name: `p_mild2severe`

f(x) Expression: `rateToProb(0.3,3,1)`

Progress

`p_mild2severe`

Run The
Model Again



I remember from looking at Mild that $p_{\text{disease_death_mild}}$ was higher than p_{die} for most ages

age	p_die
30.0	0.00132...
31.0	0.00163...
32.0	0.00174...
33.0	0.00184...
34.0	0.00195...
35.0	0.00206...
36.0	0.00219...
37.0	0.00232...
38.0	0.00245...
39.0	0.00259...
40.0	0.00272...
41.0	0.00290...
42.0	0.00309...
43.0	0.00327...
44.0	0.00346...
45.0	0.00364...
46.0	0.00392...
47.0	0.00420...
48.0	0.00447...
49.0	0.00475...
50.0	0.00503...
51.0	0.00547...
52.0	0.00591...

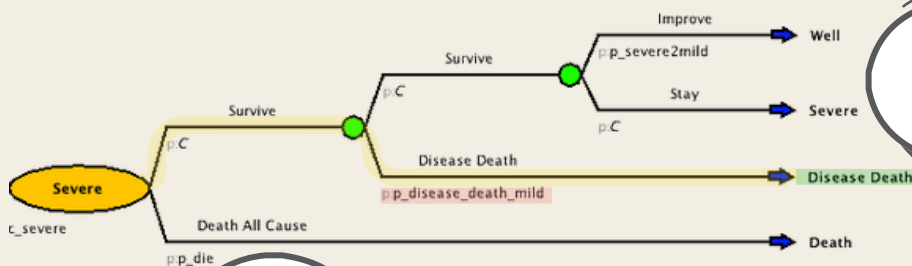
"Mild cases have a 1.4% annual probability of dying"

"Severe cases have a 4.21% annual probability of dying"

Knowing this, I am surprised to see more all cause deaths than disease deaths

I know that the Mild state is correct. So let me check the Severe state

Looking at the flow, I see that there is a branch correctly leading to the end state "Disease Death"



However, I noticed that the probability is $p_{\text{disease_death_mild}}$

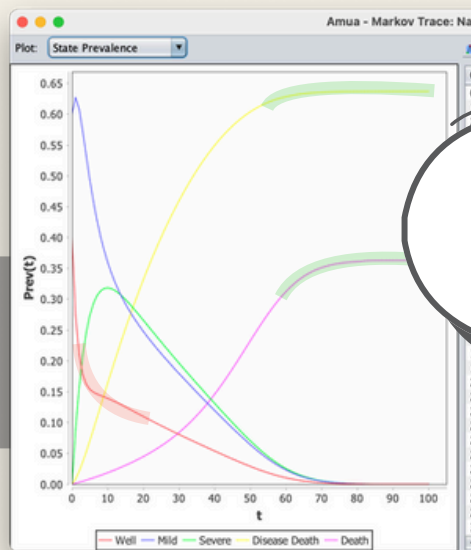
Fix #7

Change mild to severe

Disease Death
 $p_{\text{p_disease_death_mild}}$

Disease Death
 $p_{\text{p_disease_death_severe}}$

Run The
Model Again



Okay, this is looking really good. I'm making a mental note about the wiggle in Well but I want to move on to outcomes since i have reviewed a lot of the structure.

A quick look at the results shows me a glaring issue.

Strategy	Utility	Cost	Utility (Dis)	Cost (Dis)
Treatment	23921.8	5.93E8	23921.8	3.56E8

"3% discounting rate for both costs and utility"

The cost is being discounted but "Utility" and "Utility (Dis)" are the same.

Let me check the model properties

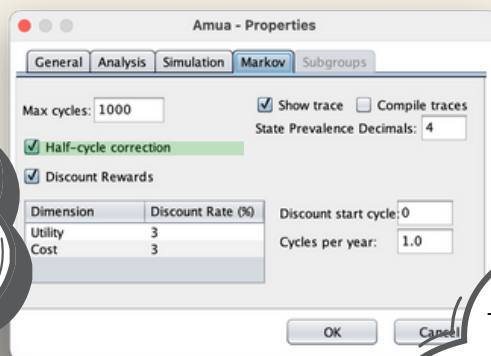
Remember that discounting should be applied to both to utility and costs. They do not have to be the same percentage though.

Fix #8
Discount Utility

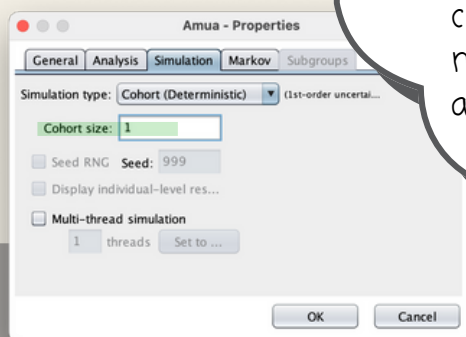
Dimension	Discount Rate (%)
Utility	0
Cost	3

Dimension	Discount Rate (%)
Utility	3
Cost	3

Well here, I also confirmed that half-cycle correction was used



To help make the results more understandable to me, I am going to use a cohort size = 1. This gives me individual-level cost and utility



Run The
Model Again

Strategy	Utility	Cost	Utility (Dis)	Cost (Dis)
Treatment	23.92	\$593,444	14.00	\$355,617

Looking at the numbers on the individual-level gives me a much better understanding of how realistic they seem

Utility tells me the additional life years that a person will have.

With an initial age of 30. People living to 54ish on average feels good to me.

I would be concerned if average age of death was super high (like 130) or super low (like 31).

Now I want to look at the cost.

Strategy	Utility	Cost	Utility (Dis)	Cost (Dis)
Treatment	23.92	\$593,444	14.00	\$355,617

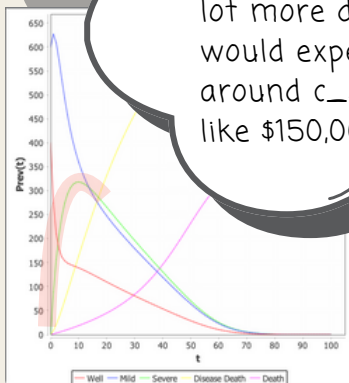
"Severe cases, a \$9,000 trip per year"

"Patients moving from mild to severe, have a one-time treatment at progression of ... \$35,000."

"Mild cases, a \$3,000 trip per year"

Knowing that most costs are low, I want to see in the trace if a lot of people are in severe

Looking at this, I see that around cycle 20 there are a lot more deaths than illness. I would expect costs to be around $c_{\text{sick}} * 20$. Expecting like \$150,000 - \$200,000.

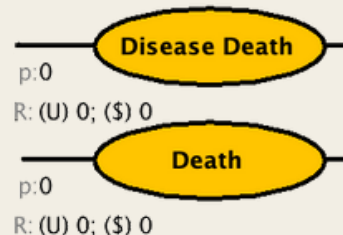


I am going to check the costs on each state.

Since you are already looking at costs. It's a good time to just quickly check the utility values as well, even if the results feel realistic

A quick check of the death states

- ✓ Utility is 0
- ✓ Cost is 0



Well

- ✓ Utility is u_{healthy}
- ✓ $u_{\text{healthy}} = 1$
- ✓ Cost is c_{healthy}
- ✓ $c_{\text{healthy}} = 0$

"Utility weight
for Mild BYE is
0.88 "

Mild

- ✓ Utility is u_{mild}
- ✓ $u_{\text{mild}} = 0.88$
- ✗ Cost is c_{mild}
- ✓ $c_{\text{mild}} = 3,000$

Well

(U) u_{healthy}
(\$) c_{healthy}

Mild

(U) u_{mild}
(\$) $c_{\text{mild}} + c_{\text{treatment}}$

Severe

(U) u_{severe}
(\$) c_{severe}

The cost of
treatment is being
applied EVERY CYCLE
to ALL individuals in
the Mild state

Patients moving from
mild to severe, have
a one-time treatment
at progression of ...
\$35,000."

But we want a
one-time cost at
progression

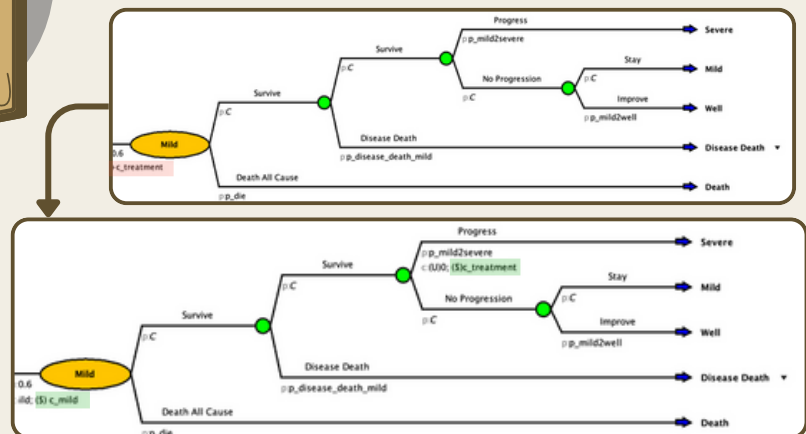
Fix #9

Move cost of
treatment

Severe

- ✓ Utility is u_{severe}
- ✓ $u_{\text{mild}} = 0.65$
- ✓ Cost is c_{severe}
- ✓ $c_{\text{severe}} = \$9,000$

"Utility weight
for Severe BYE
is 0.65 "





Strategy	Utility	Cost	Utility (Dis)	Cost (Dis)
Treatment	23.92	\$179,520	14.00	\$102,375

This price seems more reasonable based on most people getting sick and living to around 50.

Now that the results are looking good. I want to revisit the weird bump I saw in Well in the Markov Trace

I think I want to start by checking all the arms that lead to Well since I have already checked everything in the Well branch

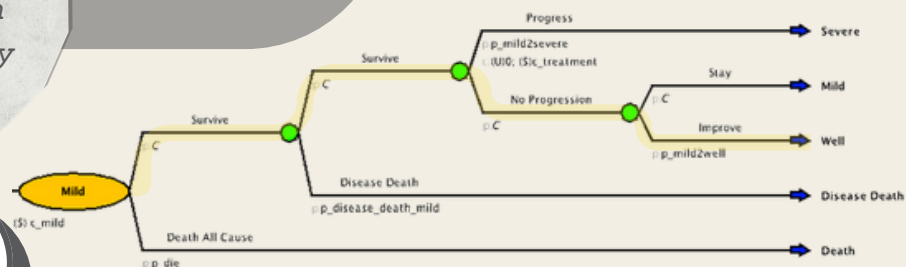


This should only be from the Mild State

I have already checked the two death probabilities and $p_{\text{mild2severe}}$

"Mild *BYE* can fully recover and return to the Well state, with an annual probability of 7%"

✓ $p_{\text{mild2well}} = 0.07$



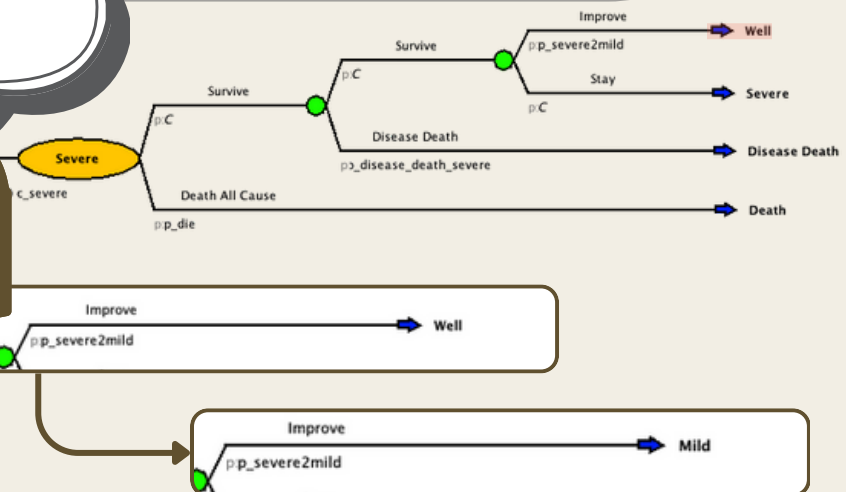
This all looks correct so now I need to figure out why this bump is happening

Let me just see if the Severe state somehow leads into Well.

Ahhhh, there is the issue. When I look at "Improve" it is going to well instead of "Mild"

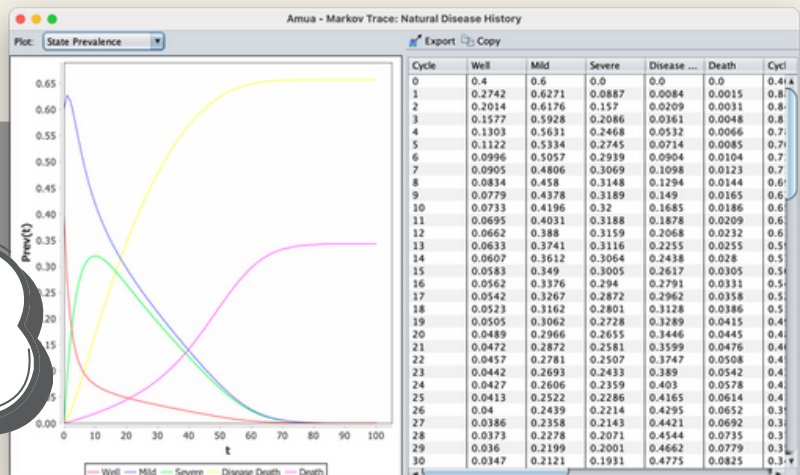
Fix #10

Change Well to Mild



Run The Model Again

We're looking great!



Strategy	Utility	Cost	Utility (Dis)	Cost (Dis)
Treatment	23.08	\$189,585	13.65	\$108,127

Now i'm going to look at the resources that were provided to see if I got similar results

Strategy	Cost (Dis)	Utility (Dis)	Inc. Cost	Inc. Utility	ICER
No Treatment	\$78,620	11.56	-	-	-
Screen then Treat	\$103,245	10.18	\$24,625	-1.38	-\$17,844.20
Treatment	\$107,989	13.45	\$4,744	3.27	\$1,450.76

Oh No! Right away I notice a negative ICER.

Do not report negative ICERs. They are ambiguous; the strategy can be dominated or dominant. Report the status instead.

Strategy	Cost (Dis)	Utility (Dis)	Inc. Cost	Inc. Utility	ICER	Status
No Treatment	\$78,620	11.56	-	-	-	
Treatment	\$107,989	13.45	\$29,369	1.89	\$15,539.15	
Screen then Treat	\$103,245	10.18	\$24,625	-1.38	--	Dominated

Now, I can actually compare.

My results are similar and it makes sense that we did not consider screening.

"Treatment:
Cost (Dis) = \$108,127
Utility(Dis) = 13.65"

The second resource also has an Incremental CEA table for me to review.

Strategy	Cost (Dis)	Utility (Dis)	Inc. Cost	Inc. Utility	ICER
No Treatment	\$86,340	12.35	-	-	-
Screen then Treat	\$107,650	13.79	\$21,310	1.44	\$14,799
Treatment	\$109,145	14.08	\$22,805	1.73	\$13,182

Seems like they also found "Screen then treat to be a viable strategy. Let me look at this closer.

Huh, the incremental cost from "Screen then Treat" to "Treat is wrong. I need to look into this.

Seems like both strategies are compared to "No Treat"

Strategy	Cost (Dis)	Utility (Dis)	Inc. Cost	Inc. Utility	ICER
No Treatment	\$86,340	12.35	-	-	-
Screen then Treat	\$107,650	13.79	\$21,310	1.44	\$14,799
Treatment	\$109,145	14.08	\$1,495	0.29	\$5,155

! Economic evaluation uses marginal (not absolute) contribution. This helps remove dominated strategies and allows us to identify the best value for cost.

Okay, now when I correctly calculate the incremental CEA, I see the screening strategy dominated again.

Strategy	Cost (Dis)	Utility (Dis)	Inc. Cost	Inc. Utility	ICER	Status
No Treatment	\$86,340	12.35	-	-	-	
Treatment	\$109,145	14.08	\$22,805	1.73	\$13,182	
Screen then Treat	\$107,650	13.79				Dominated

"Treatment:
Cost (Dis) = \$108,127
Utility(Dis) = 13.65"

Once again, we have similar findings.

Key takeaways

Effective model QA:

- **Run automated checks** - Use software validation features first
- **Review Markov trace visually** - Look for unexpected patterns
- **Examine outputs** - Check if results are in plausible range
- **Trace backwards from anomalies** - When something looks wrong, follow the logic chain back to find the error
- **Verify against source materials** - Cross-check key parameters and structure against background documents
- **Double check your model** - Check structure, settings, parameters, costs, AND interpretation

1. Model Structure Integrity

Absorbing States Must Absorb

- **Key Learning:** End states (like death) should have a 100% probability of remaining in that state
- **Why it matters:** Without this, your model suggests people can leave death or other terminal states
- **How to check:** Absorbing state populations should never decrease in the Markov trace

Verify All State Transitions Match Inputs (& Reality)

- **Key Learning:** Every pathway between states should reflect actual transitions
- **Why it matters:** Impossible transitions create unrealistic health trajectories
- **How to check:** Look for unexpected spikes or drops in Markov trace

Use Causal (Not Predictive) Parameters

- **Key Learning:** Model parameters should represent causal effects of interventions, not observational associations
- **Why it matters:** Treating a predictor of disease doesn't prevent disease unless that predictor is on the causal pathway
- **How to check:** Ask "Does this parameter capture the causal effect of the intervention?" not "Is this parameter associated with better outcomes?"

2. Time Horizon and Model Settings

Match Model Horizon to Research Question

- **Key Learning:** Termination conditions should allow the model to answer your research question fully
- **Why it matters:** Cutting off too early underestimates long-term costs and benefits
- **How to check:** For lifetime models, ensure nearly all cohort reaches absorbing states; verify cycle count in the Markov trace matches intended time period

Apply Discounting to All Relevant Outcomes

- **Key Learning:** Both costs and health outcomes typically need discounting
- **Why it matters:** Missing discounting on any dimension biases your cost-effectiveness estimates
- **How to check:** Confirm that discounted and undiscounted columns differ.

3. Parameter Values and Data Inputs

Verify Table Lookups Use Correct Index Variables

- Key Learning: When using lookup tables (e.g., mortality, utility), ensure the index matches what you intend (age vs. cycle number)
- Why it matters: Wrong indexing applies inappropriate values throughout the model
- How to check: Trace a few values manually; check if time-varying parameters behave as expected

Watch for Data Entry Errors

- Key Learning: Simple typos, misattributed values, and misplacement in models can dramatically change results
- Why it matters: Small input errors cascade into large output errors
- How to check: Double check every parameter in your model.

Never Mix Rates and Probabilities

- Key Learning: Rates and probabilities are different mathematical constructs requiring conversion
- Why it matters: Rates and probabilities can not be mathematically combined.
- How to check: Confirm conversions applied when needed

4. Cost and Utility Assignments

Distinguish Between State Costs and Event Costs

- Key Learning: Ongoing state costs (per cycle) differ from one-time event costs (at transition)
- Why it matters: Misclassification dramatically inflates or deflates total costs
- How to check: Verify costs align with clinical practice (annual treatment vs one-time intervention)

5. Results Interpretation and Reporting

Never Report Negative ICERs

- Key Learning: Negative ICERs are ambiguous; they could mean "dominant" or "dominated"
- Why it matters: Readers cannot interpret cost-effectiveness.
- How to check: When ICER is negative, determine if more costly/less effective (dominated) or less costly/more effective (dominant) and report that status.

Always Use Incremental (Not Absolute) Analysis

- Key Learning: Compare each strategy to the next-best alternative, not all to baseline
- Why it matters: Absolute comparisons miss dominated strategies and don't show marginal value
- How to check: Verify incremental costs/effects calculated sequentially after ranking by cost

Remember: Models that "run" are not necessarily correct. Verification requires multiple forms of evidence that your implementation matches your conceptual model.